

Step 1 — Missing Value Analysis

The screenshot displays the Microsoft Excel interface with the following components:

- Formula Bar:** Shows the formula `=COUNTBLANK(F2:F35)` in cell I3.
- Worksheet:** The 'Clean-Dataset' tab is active. The data is organized in columns A through I, with rows 1 through 20. The columns are: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Missing Value Analysis, Before Cleaning, and After Cleaning.
- Missing Value Analysis:** The 'Missing Value Analysis' column (F) contains text indicating missing values: 'Missing Price Values' in row 2 and 'Missing Category Values' in row 3. All other cells in this column are blank.
- Before Cleaning:** The 'Before Cleaning' column (H) contains the count of missing values: 3 in row 2 and 4 in row 3. All other cells are blank.
- After Cleaning:** The 'After Cleaning' column (I) shows the corrected values: 3 in row 2 and 0 in row 3. All other cells are blank.

	A	B	C	D	E	F	G	H	I	J	K
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Missing Value Analysis	Before Cleaning	After Cleaning		
1											
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	Missing Price Values	3	3		
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Missing Category Values	4	0		
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen					
5	11-APR-US	Smartphone	Samsung	900	25	Electronics					
6	22-MAY-US	Backpack	North Face	70	20	Accessories					
7	07-JUN-UK	Headphones	Sony		45	Electroni					
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion					
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen					
10	05-SEP-UK	Tablet	Apple	500	50	Electronics					
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor					
12	17-JUN-IN	Laptop	HP	950	25	Electronics					
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion					
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen					
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni					
16	16-APR-ES	Headphones	Bose	250	20	Electroni					
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories					
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni					
19	27-JAN-IT	Laptop	Asus	980	10	Electronics					
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion					

Observation

Identified missing values in the dataset. The 3 missing Price values were kept blank, while the missing Category values were corrected.

Step 2 — Product Name & Category Corrections

The screenshot displays the Microsoft Excel interface with a dataset of products. The 'Find and Replace' dialog is open, and a 'Microsoft Excel' message box indicates that 6 replacements were made. The dataset is organized into columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Missing Value Analysis, Before Cleaning, and After Cleaning. The 'Clean-Dataset' tab is active, showing the corrected data.

	A	B	C	D	E	F	G	H	I	J	K
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Missing Value Analysis	Before Cleaning	After Cleaning		
1	28-JAN-US	Laptop	Dell	1000	30	Electronics	Missing Price Values	3	3		
2	15-FEB-US	Sneakers	Nike	80	15	Fashion	Missing Category Values	4	0		
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen					
4	11-APR-US	smartphone	Samsung	900	25	Electronics					
5	22-MAY-US	Backpack	North Face	70	20	Accessories					
6	07-JUN-UK	Headphones	Sony		45	Electroni					
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion					
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen					
9	05-SEP-UK	Tablet	Apple	500	50	Electronics					
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor					
11	17-JUN-IN	Laptop	HP	950	25	Electronics					
12	25-NOV-AU	Sneakers	Adidas	90	40	Fashion					
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen					
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni					
15	16-APR-ES	headphones	Bose	250	20	Electroni					
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories					
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni					
18	27-JAN-IT	Laptop	Asus	980	10	Electronics					
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion					
20											

Observation

Corrected inconsistent Product Name and Category values to maintain clean and consistent data.

Step 3 — Duplicate Removal

The screenshot shows the Microsoft Excel interface with the 'Remove Duplicates' dialog box open. The dialog box displays the message: '3 duplicate values found and removed; 31 unique values remain. Note that counts may include empty cells, spaces, etc.' The background shows a data table with the following columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Missing Value Analysis, Before Cleaning, and After Cleaning. The data table contains 20 rows of product information.

	A	B	C	D	E	F	G	H	I	J	K
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Missing Value Analysis	Before Cleaning	After Cleaning		
1	28-JAN-US	Laptop	Dell	1000	30	Electronics	Missing Price Values	3	3		
2	15-FEB-US	Sneakers	Nike	80	15	Fashion	Missing Category Values	4	0		
3	03-MAR-US	Coffee Maker	Keurig	130							
4	11-APR-US	Smartphone	Samsung	900							
5	22-MAY-US	Backpack	North Face	70							
6	07-JUN-UK	Headphones	Sony								
7	19-JUL-UK	T-shirt	Adidas	30							
8	23-AUG-UK	Blender	Ninja	90							
9	05-SEP-UK	Tablet	Apple	500							
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor					
11	17-JUN-IN	Laptop	HP	950	25	Electronics					
12	25-NOV-AU	Sneakers	Adidas	90	40	Fashion					
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen					
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics					
15	16-APR-ES	Headphones	Bose	250	20	Electronics					
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories					
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics					
18	27-JAN-IT	Laptop	Asus	980	10	Electronics					
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion					
20											

Observation

Duplicate records were identified and removed to improve data consistency and avoid repeated records.

Step 4 — Product ID Split & Product Brand

The screenshot displays an Excel spreadsheet with the following data:

	B	C	D	E	F	G	H	I	J	K	L	M
	Product Name	Brand Name	Price (\$)	Quantity	Category	Missing Value Analysis	Before Cleaning	After Cleaning	Manufacturing Date	Country Code	Product Brand	
1	Laptop	Dell	1000	30	Electronics	Missing Price Values	3	3	28-JAN	US	Dell - Laptop	
2	Sneakers	Nike	80	15	Fashion	Missing Category Values	4	0	15-FEB	US	Nike - Sneakers	
3	Coffee Maker	Keurig	130	40	Kitchen				03-MAR	US	Keurig - Coffee Maker	
4	Smartphone	Samsung	900	25	Electronics				11-APR	US	Samsung - Smartphone	
5	Backpack	North Face	70	20	Accessories				22-MAY	US	North Face - Backpack	
6	Headphones	Sony		45	Electronics				07-JUN	UK	Sony - Headphones	
7	T-shirt	Adidas	30	5	Fashion				19-JUL	UK	Adidas - T-shirt	
8	Blender	Ninja	90	35	Kitchen				23-AUG	UK	Ninja - Blender	
9	Tablet	Apple	500	50	Electronics				05-SEP	UK	Apple - Tablet	
10	Hiking Boots	Timberland	130	10	Outdoor				14-OCT	UK	Timberland - Hiking Boots	
11	Laptop	HP	950	25	Electronics				17-JUN	IN	HP - Laptop	
12	Sneakers	Adidas	90	40	Fashion				25-NOV	AU	Adidas - Sneakers	
13	Coffee Maker	Nespresso	120	35	Kitchen				08-DEC	DE	Nespresso - Coffee Maker	
14	Smartwatch	Fitbit	150	15	Electronics				18-FEB	CA	Fitbit - Smartwatch	
15	Headphones	Bose	250	20	Electronics				16-APR	ES	Bose - Headphones	
16	Laptop Bag	Samsonite	50	35	Accessories				21-AUG	CA	Samsonite - Laptop Bag	
17	Smartwatch	Huawei	160	15	Electronics				20-AUG	CN	Huawei - Smartwatch	
18	Laptop	Asus	980	10	Electronics				27-JAN	IT	Asus - Laptop	
19	Sunglasses	Oakley	150	15	Fashion				01-MAR	UK	Oakley - Sunglasses	
20												

Observation

Product ID was split into Manufacturing Date and Country Code. Product Brand was created by combining Brand Name and Product Name.

Step 5 — Price Currency Formatting

The screenshot displays the Microsoft Excel interface. The ribbon at the top includes the 'Number' tab, which is active, showing the 'Currency' dropdown menu. The formula bar shows the value '1000'. The worksheet contains a table with 13 columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Missing Value Analysis, Before Cleaning, After Cleaning, Manufacturing Date, Country Code, and Product. The 'Price (\$)' column contains values formatted as currency, such as \$1,000.00, \$80.00, \$130.00, \$900.00, \$70.00, \$30.00, \$90.00, \$500.00, \$130.00, \$950.00, \$90.00, \$120.00, \$150.00, \$250.00, \$50.00, \$160.00, \$980.00, and \$150.00. The 'Before Cleaning' and 'After Cleaning' columns contain numerical values. The 'Manufacturing Date' column contains dates in YYYY-MM format. The 'Country Code' column contains two-letter codes. The 'Product' column contains the product name and brand name, such as 'Dell - L', 'Nike - S', 'Keurig - Co', 'Samsung - S', 'North Face -', 'Sony - Hea', 'Adidas -', 'Ninja - E', 'Apple -', 'Timberland - I', 'HP - L', 'Adidas - S', 'Nespresso - C', 'Fitbit - Sm', 'Bose - Hea', 'Samsonite -', 'Huawei - S', 'Asus - I', and 'Oakley - S'.

	A	B	C	D	E	F	G	H	I	J	K	L
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Missing Value Analysis	Before Cleaning	After Cleaning	Manufacturing Date	Country Code	Product
1	28-JAN-US	Laptop	Dell	\$1,000.00	30	Electronics	Missing Price Values	3	3	28-JAN	US	Dell - L
2	15-FEB-US	Sneakers	Nike	\$80.00	15	Fashion	Missing Category Values	4	0	15-FEB	US	Nike - S
3	03-MAR-US	Coffee Maker	Keurig	\$130.00	40	Kitchen				03-MAR	US	Keurig - Co
4	11-APR-US	Smartphone	Samsung	\$900.00	25	Electronics				11-APR	US	Samsung - S
5	22-MAY-US	Backpack	North Face	\$70.00	20	Accessories				22-MAY	US	North Face -
6	07-JUN-UK	Headphones	Sony		45	Electronics				07-JUN	UK	Sony - Hea
7	19-JUL-UK	T-shirt	Adidas	\$30.00	5	Fashion				19-JUL	UK	Adidas -
8	23-AUG-UK	Blender	Ninja	\$90.00	35	Kitchen				23-AUG	UK	Ninja - E
9	05-SEP-UK	Tablet	Apple	\$500.00	50	Electronics				05-SEP	UK	Apple -
10	14-OCT-UK	Hiking Boots	Timberland	\$130.00	10	Outdoor				14-OCT	UK	Timberland - I
11	17-JUN-IN	Laptop	HP	\$950.00	25	Electronics				17-JUN	IN	HP - L
12	25-NOV-AU	Sneakers	Adidas	\$90.00	40	Fashion				25-NOV	AU	Adidas - S
13	08-DEC-DE	Coffee Maker	Nespresso	\$120.00	35	Kitchen				08-DEC	DE	Nespresso - C
14	18-FEB-CA	Smartwatch	Fitbit	\$150.00	15	Electronics				18-FEB	CA	Fitbit - Sm
15	16-APR-ES	Headphones	Bose	\$250.00	20	Electronics				16-APR	ES	Bose - Hea
16	21-AUG-CA	Laptop Bag	Samsonite	\$50.00	35	Accessories				21-AUG	CA	Samsonite -
17	20-AUG-CN	Smartwatch	Huawei	\$160.00	15	Electronics				20-AUG	CN	Huawei - S
18	27-JAN-IT	Laptop	Asus	\$980.00	10	Electronics				27-JAN	IT	Asus - I
19	01-MAR-UK	Sunglasses	Oakley	\$150.00	15	Fashion				01-MAR	UK	Oakley - S

Observation

Price values were formatted consistently as currency for clear and professional presentation.

Step 6 — Price Data Bars & Electronics Highlighting

The screenshot displays the Microsoft Excel interface with a dataset of 20 rows and 13 columns. The 'Price (\$)' column (D) features orange data bars representing the price of each item. The 'Category' column (F) has blue background highlighting for all 'Electronics' entries. The interface includes the standard Excel ribbon with tabs for Clipboard, Font, Alignment, Number, Styles, Cells, Editing, and Add-ins. The status bar at the bottom shows the current temperature as 33°C and the date as 19-09-2026.

	A	B	C	D	E	F	G	H	I	J	K	L
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Missing Value Analysis	Before Cleaning	After Cleaning	Manufacturing Date	Country Code	Product
1	28-JAN-US	Laptop	Dell	\$1,000.00	30	Electronics	Missing Price Values	3	3	28-JAN	US	Dell - L
2	15-FEB-US	Sneakers	Nike	\$80.00	15	Fashion	Missing Category Values	4	0	15-FEB	US	Nike - Sn
3	03-MAR-US	Coffee Maker	Keurig	\$130.00	40	Kitchen				03-MAR	US	Keurig - Co
4	11-APR-US	Smartphone	Samsung	\$900.00	25	Electronics				11-APR	US	Samsung - S
5	22-MAY-US	Backpack	North Face	\$70.00	20	Accessories				22-MAY	US	North Face -
6	07-JUN-UK	Headphones	Sony		45	Electronics				07-JUN	UK	Sony - Hea
7	19-JUL-UK	T-shirt	Adidas	\$30.00	5	Fashion				19-JUL	UK	Adidas -
8	23-AUG-UK	Blender	Ninja	\$90.00	35	Kitchen				23-AUG	UK	Ninja - E
9	05-SEP-UK	Tablet	Apple	\$500.00	50	Electronics				05-SEP	UK	Apple -
10	14-OCT-UK	Hiking Boots	Timberland	\$130.00	10	Outdoor				14-OCT	UK	Timberland - I
11	17-JUN-IN	Laptop	HP	\$950.00	25	Electronics				17-JUN	IN	HP - L
12	25-NOV-AU	Sneakers	Adidas	\$90.00	40	Fashion				25-NOV	AU	Adidas - S
13	08-DEC-DE	Coffee Maker	Nespresso	\$120.00	35	Kitchen				08-DEC	DE	Nespresso - C
14	18-FEB-CA	Smartwatch	Fitbit	\$150.00	15	Electronics				18-FEB	CA	Fitbit - Sm
15	16-APR-ES	Headphones	Bose	\$250.00	20	Electronics				16-APR	ES	Bose - Hea
16	21-AUG-CA	Laptop Bag	Samsonite	\$50.00	35	Accessories				21-AUG	CA	Samsonite -
17	20-AUG-CN	Smartwatch	Huawei	\$160.00	15	Electronics				20-AUG	CN	Huawei - S
18	27-JAN-IT	Laptop	Asus	\$980.00	10	Electronics				27-JAN	IT	Asus - I
19	01-MAR-UK	Sunglasses	Oakley	\$150.00	15	Fashion				01-MAR	UK	Oakley - S
20												

Observation

Data Bars were applied to the Price column, and Electronics category values were highlighted for easy identification.